

Robust Trust

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Outline

- 1 Model
- 2 Trust Region Strategies
- 3 Rationalizability and MVA
- 4 Binary State Analysis
- 5 Applications and Extensions

Motivation: Trusting an Expert Adviser

Canonical Problem:

- Agent receives a recommendation from an expert who observes the state
- Expert may be **aligned** (pro-agent) or **adversarial** (anti-agent)
- Agent does not know which type

Running Example:

- AI system advises on medical treatment
- Financial adviser suggests portfolio
- Consultant reports on project viability

Core Tension:

- **Aligned**: more information $\Rightarrow$ better decisions
- **Adversarial**: more trust $\Rightarrow$ greater manipulation

Research Question:

What is the agent's *optimal strategy* for using reports, given uncertainty about alignment?

Answer: A *Trust Region Strategy*—follow reports inside a “trusted” set; clip extremes via Bregman projection.

Model: Environment

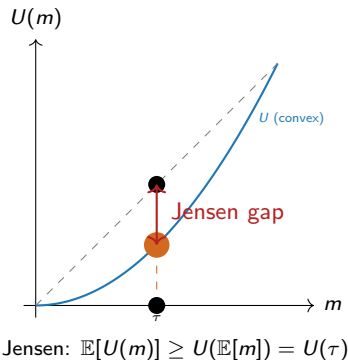
Primitives:

- Finite state space $\Omega = \{1, \dots, N\}$, prior $\tau \in \Delta(\Omega)$
- Agent has private type $\theta \in \Theta$, takes action $a \in A$
- Stage utility: $u(a, \theta, \omega)$

Agent's Value Function:

$$U(m) = \mathbb{E}_{\theta} \left[\max_a \mathbb{E}_m[u(a, \theta, \omega)] \right]$$

- First-best payoff given posterior $m \in \Delta(\Omega)$
- U is **convex** in m : $\max_a(\cdot)$ preserves convexity, expectation over θ preserves it
- Baseline (no information): $U(\tau)$



Model: Properties of the Value Function

Why Convexity Matters:

- Blackwell's theorem: aligned adviser with more information always raises $\mathbb{E}[U(m)]$
- Q dominates Q' iff $\mathbb{E}_Q[U(m)] \geq \mathbb{E}_{Q'}[U(m)]$ for all convex U
- Convexity is *necessary and sufficient* for information to be valuable

Bregman Divergence of U :

$$D_U(m, m') = U(m) - U(m') - \nabla U(m') \cdot (m - m')$$

- Gap between $U(m)$ and tangent to U at m'
- $D_U(m, m') \geq 0$; $= 0$ iff $m = m'$ (strict convexity)
- **Not symmetric:** $D_U(m, m') \neq D_U(m', m)$

Economic Interpretation of D_U :

$$D_U(m, m') = U(m) - \mathbb{E}_m[u(\hat{\sigma}(m'), \omega)]$$

where $\hat{\sigma}(m') = \operatorname{argmax}_a \mathbb{E}_{m'}[u(a, \theta, \omega)]$.

- = Payoff loss from acting on m' when truth is m
- = Adversary's gain from inducing action $\hat{\sigma}(m')$ when state is drawn from m
- Key: $D_U(m, m')$ is exactly what the adversary gains by "moving" the agent's belief from m to m'

Model: Adviser Types and Agent's Objective

Adviser: observes state ω , sends report $m \in \Delta(\Omega)$

- **Aligned** (prob α): sends true posterior; agent gets $U(\tilde{m})$
- **Adversarial** (prob $1 - \alpha$): chooses reporting strategy $\beta : \Omega \rightarrow \Delta(\Delta(\Omega))$ to minimize agent utility

Agent's strategy: $\sigma : \Delta(\Omega) \rightarrow \Delta(A)$

Worst-Case Payoff:

$$V(\sigma; \alpha) = \alpha \mathbb{E}[U(\tilde{m})] + (1 - \alpha) \inf_{\beta} \mathbb{E}_{\beta}[u(\sigma(m), \omega)]$$

Interpretation of Each Term:

- $\alpha \mathbb{E}[U(\tilde{m})]$: aligned contribution — follows Blackwell, always $\geq \alpha U(\tau)$
- $(1 - \alpha) \inf_{\beta} \dots$: adversarial floor — adversary picks worst reporting strategy β^*

Agent's Problem: $\max_{\sigma} V(\sigma; \alpha)$ **Baseline:**

$\sigma(\cdot) \equiv \hat{\sigma}(\tau)$ gives $V = U(\tau)$ for all α

Trust Region Strategies: Definition

Definition 1 (Trust Region Strategy, TRS)

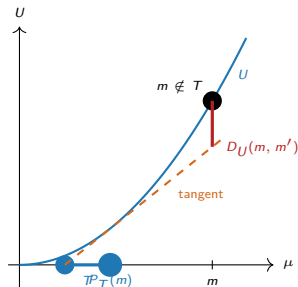
A TRS is parameterized by a closed connected set $T \subseteq \Delta(\Omega)$. Given report m , act on the *Bregman projection* $P_T(m) = \operatorname{argmin}_{m' \in T} D_U(m, m')$: take action $\hat{\sigma}(P_T(m)) = \operatorname{argmax}_a \mathbb{E}_{P_T(m)}[u(a, \theta, \omega)]$.

Behavior by region:

- $m \in T$: $P_T(m) = m$; act on m directly (full trust)
- $m \notin T$: clip to nearest $m' \in T$ in Bregman distance; act on m'

Interpretation:

T = set of “plausible” beliefs the agent accepts.
Extreme reports are discounted to the nearest trusted belief.



Bregman Divergence: Key Properties

Properties of $D_U(m, m')$:

1. **Non-negative:** $D_U(m, m') \geq 0$; $= 0$ iff $m = m'$
2. **Asymmetric:** $D_U(m, m') \neq D_U(m', m)$ in general
3. **Convex in m :** for fixed m' , $m \mapsto D_U(m, m')$ is convex
4. **Generalized Pythagoras:** for $m^* = P_T(m)$ and any $m' \in T$,

$$D_U(m, m') \geq D_U(m, m^*) + D_U(m^*, m')$$

5. **Three-point identity:**

$$D_U(m, m') = D_U(m, m^*) + D_U(m^*, m') + [\nabla U(m') - \nabla U(m^*)] \cdot (m - m^*)$$

Projection Minimality:

$P_T(m) = \operatorname{argmin}_{m' \in T} D_U(m, m')$ satisfies:

$$D_U(m, P_T(m)) \leq D_U(m, m') \quad \forall m' \in T$$

Why Bregman (not Euclidean)?

- Euclidean distance ignores curvature of U
- Bregman distance accounts for how costly deviations from m' are given U 's geometry
- $D_U(m, P_T(m)) = \text{exact payoff loss from being forced to act on } P_T(m) \text{ instead of } m$
- No other projection metric gives this exact interpretation

Adversary's Optimal Reply to a TRS

Setup: Agent uses TRS with trust region T ;
adversary observes state ω and sends report m .

Agent's action given m : $\hat{\sigma}(P_T(m))$

Adversary's gain from sending m at state ω :

$$U(P_T(m)) - \mathbb{E}_\omega[u(\hat{\sigma}(P_T(m)), \omega)] = D_U(\delta_\omega, P_T(m))$$

- $\equiv$ payoff loss from acting on $P_T(m)$ instead of truth δ_ω
- Adversary maximizes this over $m \in \Delta(\Omega)$

Adversary's Optimal Strategy:

- Best reply:
 $m^* = \operatorname{argmax}_{m \notin T} D_U(\delta_\omega, P_T(m))$
- Adversary sends a report *on or beyond the boundary* ∂T
- Reports inside T are projected to themselves: no gain from them
- Reports far outside T are projected to boundary of T : adversary picks boundary point causing most harm

Binary state: Adversary's best reply is $m^* \in \{0, 1\}$ (extreme beliefs); clips to $\underline{\mu}$ or $\bar{\mu}$.

Theorem 1: Trust Region Strategies Are Optimal

Theorem (Theorem 1)

For any $\alpha \in [0, 1]$, any optimal agent strategy σ^* is equivalent to a TRS with some connected trust region $T^* \subseteq \Delta(\Omega)$.

Meaning:

- WLOG restrict to TRS: the TRS class is *without loss of generality*
- Optimal agent clips reports to T^* via Bregman projection
- Connectedness is necessary: disconnected T can always be improved

Why This Is Non-Trivial:

- Clipping reports seems like it discards information
- But: adversary exploits *any* non-clipped region
- Bregman projection is the unique clipping rule that minimizes the adversary's maximum gain (not Euclidean, not other metrics)
- Connected T eliminates exploitable “gaps”

Proof of Theorem 1

Proof Sketch.

Fix any optimal σ^* . We construct a TRS with weakly higher V .

Step 1 — Bregman projection minimizes adversarial gain. For any strategy σ , let $\psi(m) \in \Delta(\Omega)$ be the “effective belief” such that $\sigma(m) = \hat{\sigma}(\psi(m))$. The adversary’s gain from sending report m equals

$$D_U(m, \psi(m)) = U(m) - \mathbb{E}_m[u(\hat{\sigma}(\psi(m)), \omega)].$$

Setting $T = \overline{\text{range}(\psi)}$ and replacing ψ with the Bregman projection P_T : since $D_U(m, P_T(m)) \leq D_U(m, \psi(m))$ for all m (by minimality of P_T), the adversary gains *weakly less* under P_T . Aligned gain is unchanged (aligned posteriors are in T). So $V(\sigma_{P_T}) \geq V(\sigma^*)$.

Step 2 — Connectedness. If T has a gap G , for any $m \in G$ the adversary sends m and the agent projects to the nearest component boundary, which may be far from m . Taking $T^{**} = \text{conv-hull}(T)$ fills G : the Bregman projection onto T^{**} is (weakly) closer to each $m \in G$ than projection onto T , so

$D_U(m, P_{T^{**}}(m)) \leq D_U(m, P_T(m))$. Adversarial gain weakly falls; aligned gain unchanged. Hence $V(\sigma_{T^{**}}) \geq V(\sigma_T)$.

Taking $T^* = T^{**}$ gives $\sigma^* \equiv \text{TRS}(T^*)$. □

Definition 2 (Robust Rationalizability)

A strategy σ is *robustly rationalizable* if there exist $\alpha' \in [0, 1]$ and adversarial strategy β' such that $\sigma \in \operatorname{argmax}_{\sigma'} V(\sigma'; \alpha')$ given adversary β' .

Interpretation:

- σ is a best response for the agent under *some* consistent belief about alignment
- Agent need not know the true α : if their belief α' rationalizes σ , then σ is self-enforcing
- “Robust”: holds for beliefs about adversary strategy, not just about α

Contrast:

- **Optimal** σ^* : best response at the *true* α
- **Rationalizable** σ : best response at *some* belief α'

Theorem 2 says these coincide!

Any optimal strategy at true α is also rationalized by (α, β^*) where β^* is the adversary's best reply.

No commitment to a mechanism required.

Theorem 2: No Commitment Needed

Theorem (Theorem 2 — Robust Rationalizability)

The following are equivalent for any agent strategy σ^* :

- (i) σ^* **maximizes** $V(\sigma; \alpha)$ over all σ (optimal)
- (ii) σ^* is **robustly rationalizable** (no commitment)
- (iii) (σ^*, β^*) forms a **saddle point**: $\max_{\sigma} \min_{\beta} V = \min_{\beta} \max_{\sigma} V$ (minimax)

Practical Consequence:

- Agent can compute optimal TRS, announce nothing, and simply follow it
- Adversary cannot do better by observing the agent's strategy ex-ante
- Order of commitment irrelevant: agent can move first or second

Contrast with Mechanism Design:

- Standard persuasion/cheap talk: sender often benefits from commitment
- Here: agent (receiver) needs no commitment
- The asymmetry arises because TRS is already a best response, not just optimal by threat

Proof of Theorem 2

Proof.

Step 1 (Minimax Theorem): The payoff $V(\sigma, \beta) = \alpha \mathbb{E}_{id, \sigma}[u] + (1 - \alpha) \mathbb{E}_{\beta, \sigma}[u]$ is bilinear in (σ, β) . The strategy spaces Σ and $\mathcal{B}$ are compact and convex. By von Neumann's minimax theorem:

$$\max_{\sigma} \min_{\beta} V(\sigma, \beta) = \min_{\beta} \max_{\sigma} V(\sigma, \beta),$$

so a saddle point (σ^*, β^*) with $V(\sigma^*, \beta) \leq V(\sigma^*, \beta^*) \leq V(\sigma, \beta^*)$ for all (σ, β) exists. This proves (iii).

Step 2 (Equivalences):

- (i) $\Leftrightarrow$ (iii): σ^* maximizes $\min_{\beta} V(\sigma, \beta) = V(\sigma; \alpha)$ iff (σ^*, β^*) is a saddle point.
- (i) $\Rightarrow$ (ii): At the saddle point, $\sigma^* = \operatorname{argmax}_{\sigma} V(\sigma, \beta^*)$, so σ^* is a best response to β^* . Hence σ^* is rationalized by beliefs (α, β^*) .
- (ii) $\Rightarrow$ (i): If σ^* is a best response under $(\alpha' = \alpha, \beta')$, then $\sigma^* \in \operatorname{argmax}_{\sigma} V(\sigma; \alpha)$. □



Minimal Viable Alignment

Definition 3 (Minimal Viable Alignment)

$$\text{MVA}(\tau) = \inf \{ \alpha \in [0, 1] : \max_{\sigma} V(\sigma; \alpha) > V(\hat{\sigma}(\tau); \alpha) \}$$

The minimum alignment probability for which advice *strictly improves* over ignoring the adviser.

Bounds:

- $\text{MVA}(\tau) \in [1/N, 1/2]$
- Upper bound $1/2$: adversary can always match aligned benefit at $\alpha = 1/2$, so advice never strictly helps unless $\alpha > 1/2$
- Lower bound $1/N$: with N states, rich information structure enables advice even at low α

Interpretation:

- $\alpha > \text{MVA}$: follow adviser (positive net value)
- $\alpha \leq \text{MVA}$: ignore adviser (adversarial risk dominates)
- **Lower MVA** $\Leftrightarrow$ advice more robust to misalignment

Connection to Bayes persuasion:

MVA is linked to the informational complexity of the optimal experiment for (τ, U) .

Theorem 3: $MVA(\tau) = 1/R(\tau)$

Theorem (Theorem 3)

Let $R(\tau) = \# \text{support points of the optimal Bayes persuasion experiment for } (\tau, U)$. Then $MVA(\tau) = 1/R(\tau) \in [1/N, 1/2]$.

Proof Sketch.

Fix a K -support experiment $\{(m_k, w_k)\}$ with aligned gain $A_K = \sum w_k U(m_k) - U(\tau) > 0$.

1. Bregman boundary balance $\Rightarrow$ adversary's gain B_K satisfies $B_K/A_K = 1/(K-1)$.

2. MVA solves $MVA \cdot A_K = (1 - MVA) \cdot B_K$:

$$MVA = \frac{B_K/A_K}{1 + B_K/A_K} = \frac{1/(K-1)}{K/(K-1)} = \frac{1}{K}.$$

3. Optimize over K : optimal $K = R(\tau)$, so $MVA(\tau) = 1/R(\tau)$. □
-

Bounds on MVA:

- $R(\tau) \geq 2$: $MVA \leq 1/2$
- $R(\tau) \leq N$: $MVA \geq 1/N$
- $N = 2$: $R(\tau) = 2$ always, $MVA = 1/2$

Interpretation:

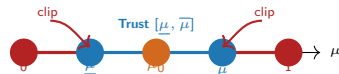
- More support points $\Rightarrow$ richer info structure $\Rightarrow$ lower MVA
- Adversary with K posteriors to exploit gains $1/(K-1)$ of aligned benefit — harder to exploit when K is large

Binary State: Setup and Corollary 2

Setting: $\Omega = \{0, 1\}$, belief $\mu = P(\omega = 1) \in [0, 1]$, prior $\mu_0 \in (0, 1)$

Corollary (Corollary 2)

In the binary-state model, any optimal strategy is equivalent to a TRS with $T = [\underline{\mu}, \bar{\mu}] \subseteq [0, 1]$.



Why an interval?

- $\Delta(\{0, 1\}) \cong [0, 1]$: beliefs are scalars
- Connected subsets of $[0, 1]$ are exactly closed intervals $[\underline{\mu}, \bar{\mu}]$
- Bregman projection clips:
 $P_{[\underline{\mu}, \bar{\mu}]}(\mu) = \max(\underline{\mu}, \min(\bar{\mu}, \mu))$
- Adversary's best reply: $\mu^* \in \{0, 1\}$ (extreme beliefs)

Proposition 1: Characterizing the Optimal Interval

Proposition (Proposition 1)

- $\alpha \leq \frac{1}{2}$: $T^* = \{\mu_0\}$ (ignore all reports)
- $\alpha > \frac{1}{2}$: $T^* = [\underline{\mu}(\alpha), \bar{\mu}(\alpha)]$, uniquely given by the lower balance condition $(1 - \alpha) D_U(0, \underline{\mu}) = \alpha \int_0^{\underline{\mu}} [U(\underline{\mu}) - U(\mu)] d\tau(\mu)$ and the symmetric condition at $\bar{\mu}$.

Balancing at Lower Boundary $\underline{\mu}$:

- **LHS**: adversary sends $\mu = 0$; gain = $D_U(0, \underline{\mu})$ weighted by $(1 - \alpha)$
- **RHS**: aligned adviser's gain from extending T below $\underline{\mu}$; weighted by α
- Optimal $\underline{\mu}$: extend T until marginal aligned gain = marginal adversarial cost

Key Observations:

- At $\alpha = 1/2$: LHS = RHS only at $\underline{\mu} = \mu_0$ (only prior is trusted)
- For $\alpha > 1/2$: expand T outward until balance is restored
- **Uniqueness**: balance condition is monotone in $\underline{\mu}$; $\underline{\mu}(\alpha)$ uniquely pinned down
- $\bar{\mu}(\alpha)$ by symmetric condition at upper boundary

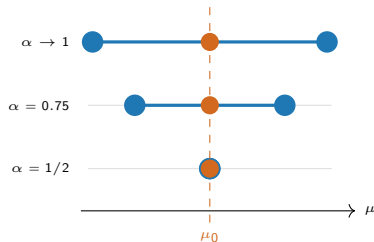
Proposition 2: Trust Expands Monotonically

Proposition (Proposition 2)

$\underline{\mu}(\alpha)$ decreases and $\bar{\mu}(\alpha)$ increases in α . At $\alpha = \frac{1}{2}$: $T = \{\mu_0\}$; as $\alpha \rightarrow 1$: $T \rightarrow [0, 1]$.

Proof Idea:

- Balancing condition:
 $(1 - \alpha)D_U(0, \underline{\mu}) = \alpha \int_0^{\underline{\mu}} [U(\underline{\mu}) - U(\mu)] d\tau$
- As α increases, RHS weight α grows relative to LHS weight $(1 - \alpha)$
- To restore balance: $\underline{\mu}$ must decrease (making LHS larger and RHS smaller)
- Monotone comparative statics: $\underline{\mu}(\alpha)$ is strictly decreasing in α



Proposition 3: Location of the Trust Region

Proposition (Proposition 3)

The trust region shifts upward (in strong set order) when U is less info-sensitive at high beliefs. Formally: if $\tau' \succ_{\text{FOSD}} \tau$, then $T^(\tau') \geq_{\text{SSO}} T^*(\tau)$.*

Proof Idea:

- **Info-sensitivity at μ :** curvature of U at μ measures how costly errors are near μ
- High curvature at low $\mu \Rightarrow$ adversary exploits downward misreports more $\Rightarrow$ trust region shifts up to reduce this exposure
- FOSD shift of τ increases the “effective curvature” at high beliefs relative to low $\Rightarrow$ SSO shift of T^* upward

Interpretation:

- Higher prior μ_0 (agent more optimistic) $\Rightarrow$ trust region is higher
- Agent already expects high state $\Rightarrow$ accepts upward reports more readily
- Asymmetry in U 's curvature $\Rightarrow$ asymmetry in trust

Example: Medical triage with $r > 1$: Under-treatment is costlier (U more curved at low μ) $\Rightarrow$ trust region shifts upward to guard against adversarially low reports.

Application: Medical Triage

Setting:

- Patient severity: $\mu = P(\text{severe})$
- AI advises on treatment intensity $a \in [0, 1]$
- Aligned AI: truthful severity reports
- Adversarial AI: may misstate severity

Value Function:

$$U(\mu) = 1 - \frac{(1 - \mu)^2}{r\mu + (1 - \mu)}$$

$r > 0$: relative cost of under-treatment vs. over-treatment; U convex in μ .

For $r = 1$, $\mu_0 = \frac{1}{2}$ (symmetric):

Explicit solution from Prop. 1 balance condition:

$$\underline{\mu}(\alpha) = \frac{\sqrt{(1 - \alpha)(1 + 2\alpha)} - (1 - \alpha)}{2\alpha}$$

$\bar{\mu}(\alpha) = 1 - \underline{\mu}(\alpha)$ by symmetry.

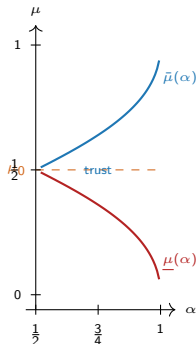
- $\underline{\mu}(1/2) = 1/2$: no trust at $\alpha = 1/2$ (= MVA)
- $\underline{\mu}(\alpha) \searrow 0$ as $\alpha \nearrow 1$

Medical Triage: Trust Region Expansion

Doctor's Optimal Policy:

- $\mu \in [\underline{\mu}(\alpha), \bar{\mu}(\alpha)]$: follow AI report
- $\mu < \underline{\mu}(\alpha)$: clip up to $\underline{\mu}(\alpha)$
(distrust low severity claims)
- $\mu > \bar{\mu}(\alpha)$: clip down to $\bar{\mu}(\alpha)$
(distrust high severity claims)

Trust range widens as α increases. At $\alpha = 1$: follow all AI reports unconditionally.



Extension: Binary Action (Proposition 4)

Setting: $A = \{0, 1\}$ (e.g., treat or don't), threshold rule: $a = 1$ iff $\mu > \mu^*$

Proposition (Proposition 4)

Let L = loss from wrongly choosing $a = 0$, G = loss from wrongly choosing $a = 1$.

$$\hat{\alpha} = \frac{\max\{L, G\}}{L + G} \geq \frac{1}{2}$$

- $\alpha > \hat{\alpha}$: $T = \Delta(\Omega)$ (trust all reports)
- $\alpha < \hat{\alpha}$: $T = \{\mu_0\}$ (trust no reports)

All-or-Nothing Trust:

- Partial trust is never optimal with binary action
- Any intermediate T allows adversary to flip the binary decision at low cost
- Symmetric ($L = G$): $\hat{\alpha} = 1/2$;
asymmetric ($L \gg G$): $\hat{\alpha} \rightarrow 1$

Proof Idea:

With binary action,

$U(\mu) = \max\{L \cdot \mu, G \cdot (1 - \mu)\}$ is piecewise linear (“tent”). Bregman projection onto any T either includes the kink or not; intermediate T always leaves a “flip” for the adversary.

Extension: Spherical Payoffs

Setting: $|\Omega| = N$; payoffs depend only on distance from reference belief b : $U(\mu) = \tilde{U}(\|\mu - b\|^2)$

Corollary (Corollary 3 / Example 1)

The optimal trust region is a ball centered at b :

$$T(\alpha) = \{\mu \in \Delta(\Omega) : \|\mu - b\|^2 \leq r^*(\alpha)^2\}$$

with $r^(\alpha)$ increasing in α .*

MVA = 1/2 for spherical payoffs (regardless of N).

- $r^*(1/2) = 0$: no trust at MVA threshold
- $r^*(1) \rightarrow \infty$ (clipped by simplex)

Why MVA = 1/2 Despite $N > 2$?

- Spherical payoff has $R(\tau) = 2$ for any N : optimal experiment has only 2 support points (center and sphere boundary), so MVA = 1/2
- Symmetric structure implies adversary can always match aligned gain at $\alpha = 1/2$

Interpretation:

Trust region geometry mirrors payoff geometry. Ball radius = willingness to accept reports far from prior b .

Main Results:

1. **Theorem 1 (TRS Optimality):**
Optimal strategy = clip to nearest trusted belief via Bregman projection onto connected T^*
2. **Theorem 2 (No Commitment):**
TRS is robustly rationalizable; saddle point holds; no precommitment required
3. **Theorem 3 (MVA):**
 $MVA(\tau) = 1/R(\tau) \in [1/N, 1/2]$; information valuable iff $\alpha > 1/R(\tau)$

Key Takeaways:

- Trust advice *within a region*; discount extremes
- Trust region *expands* monotonically with α
- Payoff curvature determines both size and location of T
- Binary action: trust is all-or-nothing

Broader Implication:

Calibrating trust to estimated alignment probability is not only normatively appealing — it is the *provably optimal* response to potential misalignment.